

APURV KUMAR SINGH

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Education

Institute of Engineering and Management, Kolkata

B.Tech in Computer Science & Engineering (AIML)

2023 – 2027

CGPA: 8.34/10

Surendranath Centenary School, Ranchi

Class XII – CBSE

2022

Percentage: 86.66%

Surendranath Centenary School, Ranchi

Class X – CBSE

2020

Percentage: 90.83%

Technical Skills

Languages: C/C++, SQL, Python

AI/ML: Computer Vision, YOLO, PyTorch, OpenCV, CNN, MATLAB, Deep Learning

Tools & Technologies: VS Code, Google Colab, Visual Paradigm, ProjectLibre, Git/Github

Cloud & Databases: AWS, MySQL

Methodologies: Agile, Scrum, SDLC

CS Fundamentals: Data Structures & Algorithms, Software Engineering, DBMS, Computer Networks

Soft Skills: Problem Solving, Teamwork, Self-learning, Communication, Time Management

Work Experience

Elevate Labs

April 2025 – May 2025

AIML Intern Certificate

Remote

- **Credit Fraud Detection:** Developed a real-time anomaly detection system using LightGBM, achieving 94% accuracy on imbalanced financial data.
- **Tech Stack:** Scikit-Learn, Seaborn, NumPy, Pandas, Matplotlib, Streamlit

Projects

Pneumonia Detection using PyTorch

- Developed a deep learning framework to automate pneumonia diagnosis from chest X-ray images, performing a comparative ablation study.
- Benchmarked a custom 3-block **CNN** against pre-trained architectures (**ResNet50**, **VGG16**, **MobileNetV2**) using transfer learning.
- Achieved best diagnostic performance with **ResNet50 (0.94 ROC-AUC)** and **MobileNetV2 (88% accuracy)**, optimizing recall for disease identification.
- **Tech Stack:** Python, PyTorch, CNN, ResNet50, VGG16, MobileNetV2, Scikit-Learn.

Dream Home – Full-Stack Real Estate App | [Live Link](#)

- Developed a comprehensive real estate listing platform with role-based dashboards for buyers, sellers, and agents.
- Implemented secure **RBAC** (Role-Based Access Control) using **JWT**, providing distinct permissions and interfaces for listing management.
- Engineered a **REST API** with **Node.js** and **MySQL** to handle full CRUD operations, multi-image uploads via Multer, and a status workflow.
- **Tech Stack:** JavaScript, Tailwind CSS, Node.js, Express.js, MySQL, JWT, Multer.

Certifications

Artificial Intelligence in Industrial and Management Engineering – IIT Kanpur | [View Certificate](#)

Cyber Security Fundamentals – University of London | [View Certificate](#)

Introduction to Applied Cryptography – University of Colorado | [View Certificate](#)

Introduction to GitHub Copilot – Microsoft | [View Certificate](#)

Co-curricular Activities

Computer Vision Hackathon Participant – [INSIGHT] | [View Certificate](#)

Google Cloud Gen AI Study Jam – Participant | [View Certificate](#)

ISTE Student Member – Indian Society for Technical Education | [View Membership](#)